

Lab 1a: Membuat AI Agent dengan watsonx.ai dan crew

Pada lab ini, kita akan membuat *Sales Analyzer Agent* menggunakan **watsonx.AI** dan **Crew AI**. Agent tersebut akan digunakan untuk melakukan analisis data dari suatu produk untuk melakukan proses *procurement*.

Notes: File yang digunakan merupakan file yang dapat di-*download* dari folder “”

Tahapan untuk membuat Agent

Step 1: Membuat “Project”

- Jika ini pertama kali anda menggunakan *account* ini. Maka anda perlu membuat *project* sebelum menggunakan *Agent Lab*. Selain itu, anda dapat menggunakan *project* yang telah ada.

IBM watsonx

2948357 - itz-watsonx-031 Dallas

Create a project

Start with a new, blank project or select from where to import an existing project.

+ New

- Local file
- Sample

Define details

Name

Lab1

Description (optional)

What's the purpose of this project?

Tags (optional)

Add tags

Add tags to make projects easier to find. To add tags, separate them with commas and press Enter.

Storage

itzc0s-69400056k5-y1q1

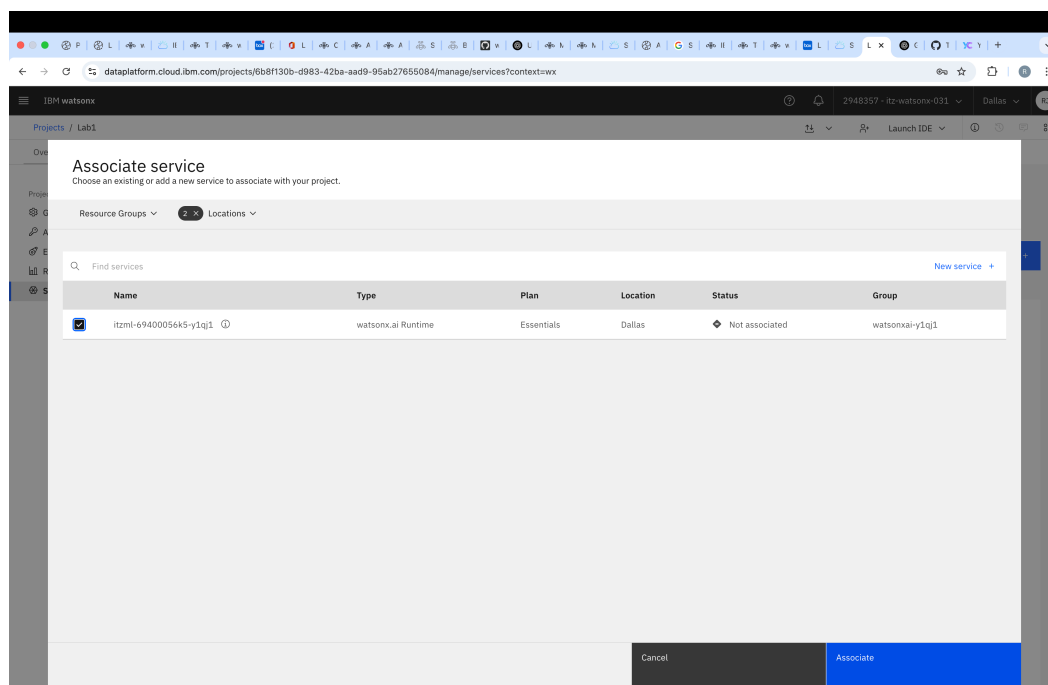
Project includes integration with [Cloud Object Storage](#) for storing project assets.

Advanced settings

Cancel Create

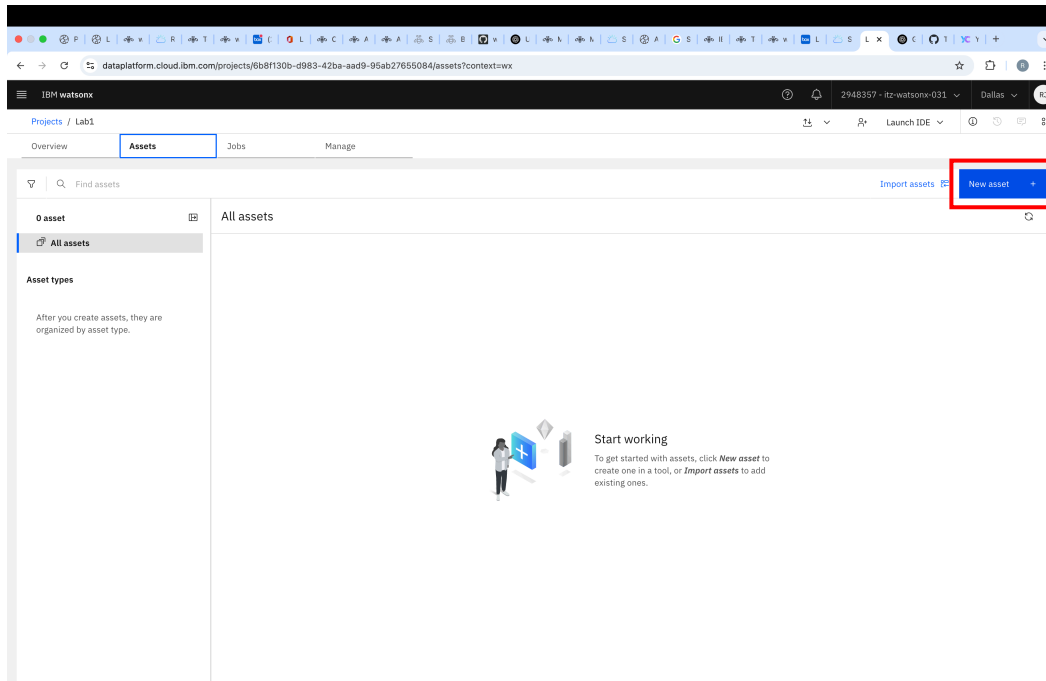
Tahap 2: Mengasosiasikan watsonx.Ai dengan watsonx.ai Runtime.

- Setelah membuat *project*, buka **Manage** → **Services & Integrations** → **Associate Service**.
- Pilih **watsonx.ai** dan asosiasikan dengan *project* yang ada.



Step 3: Add a New Asset Tahap 3: Tambahkan Asset Baru

- Pergi ke halaman *Assets Tab* dan klik *New Asset*

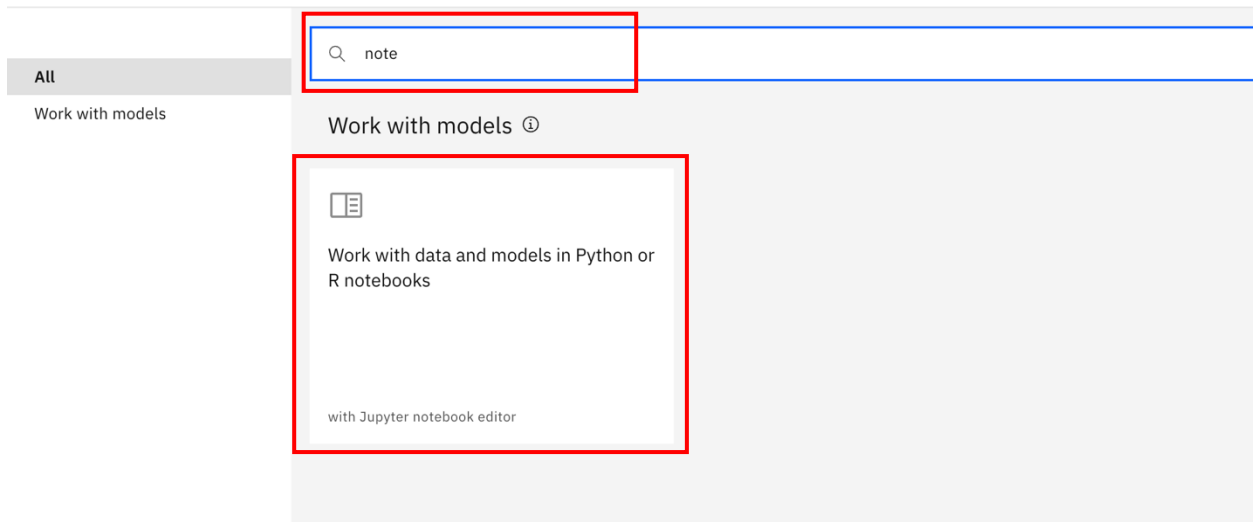


Step 4: Create the notebooks Tahap 4: Buat notebooknya.

- Cari "Work with data and models in Python or R notebooks" dan klik "Create a notebook"

What do you want to do?

Select a task based on your goal. You'll use a tool to create an asset for that goal.



Tahap 5: Upload files yang telah diberikan

- Klik “Local File.”
- Upload notebook untuk *vector index creation* menggunakan *Browse*
 - SalesAnalysis.ipynb
 - SalesAnalysis_query.ipynb
- Klik *Create* untuk membuat *notebook*

Work with data and models in Python or R notebooks

Define the details to create a notebook asset and open it in the Jupyter notebook editor tool.

The screenshot shows the 'Define details' form for creating a notebook asset. On the left, a sidebar contains options: '+ New', 'Sample', 'Local file' (highlighted with a red box), and 'URL'. The main form area is divided into two sections: 'Define details' and 'Define configuration'. Under 'Define details', there is a 'Local asset' field containing 'salesAnalysis.ipynb', a 'Name' field containing 'salesAnalysis', and a 'Description (optional)' text area with the placeholder 'What's the purpose of this notebook'. Under 'Define configuration', there is a 'Select runtime' dropdown menu set to 'Runtime 24.1 on Python 3.11 XS (2 vCPU 8 GB RAM)'. Below this, a note states: 'The selected runtime has 2 vCPU and 8 GB RAM. It consumes 1 capacity unit per hour. [Learn more](#) about capacity unit hours and watsonx.ai Studio pricing plans.' At the bottom right, there are two buttons: 'Cancel' and 'Create' (highlighted with a red box).

Work with data and models in Python or R notebooks

Define the details to create a notebook asset and open it in the Jupyter notebook editor tool.

The screenshot shows the 'Define details' form for creating a notebook asset, specifically the file upload step. The sidebar on the left has 'Local file' highlighted with a blue box. The main area features a large dashed box for file upload. Inside this box, there is an illustration of a person standing next to a blue cube with a laptop on top. To the right of the illustration, the text reads: 'Drop IPYNB file here or browse for file to upload.' Below this, a note states: 'Add a notebook (IPYNB) file that contains your notebook. You can only upload files up to 52 MB.' A blue 'Browse' button is positioned below the note. At the bottom right, there are two buttons: 'Cancel' and 'Create'.

Step 6: Configure the notebook

Tahap 6: Konfigurasi API Key pada notebook

- Masukkan *key* yang dibutuhkan untuk dapat menjalankan *notebook*.
- Buat *IBM Cloud API Key* atau dapatkan dari anggota tim lainnya dan ubah parameter “*api_key*” untuk menggunakan *LLM*. Lalu, verifikasi *api_base* sesuai dengan instans *watsonx* yang disediakan
- *DB_Password* dapat ditemukan di “*DB Connection Details.boxnote*”

```
[*]: DB_URL = "cb0cfae7-372d-4047-9bf9-882289cd9910.blijti4d0v0nkr55oei0.databases.appdomain.cloud"
DB_PORT = "32240"
DB_NAME = "marcht3db"
DB_USER = "ibm_cloud_d2c0138f_5cb3_499e_8043_6a959fb57913"
DB_PASSWORD = input("Enter password for the user:")
```

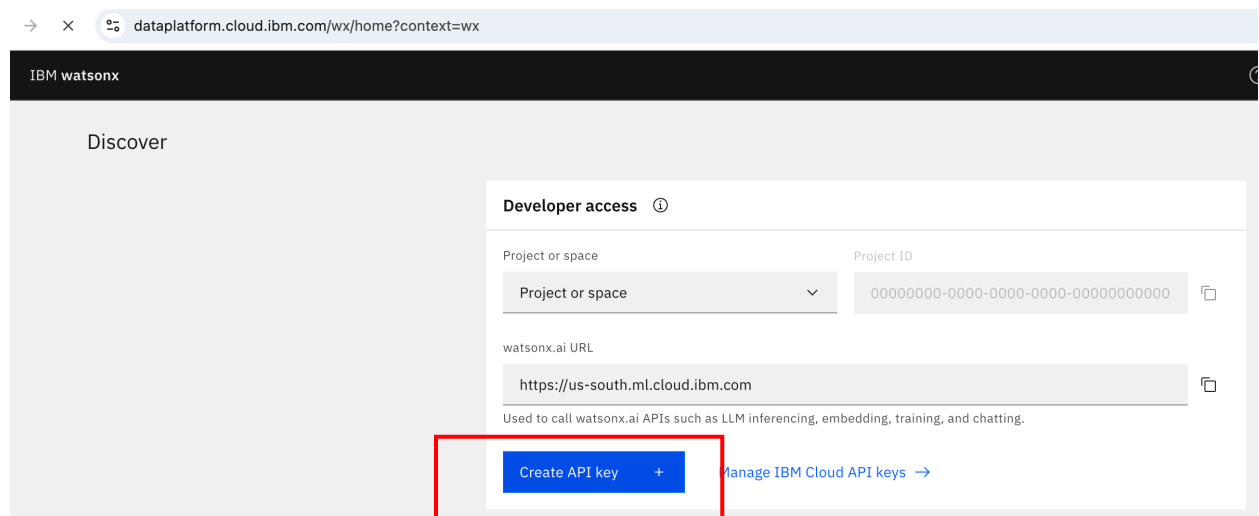
Enter password for the user:

```
[9]: nl2sql = NL2SQLTool(db_uri=f"postgresql://{DB_USER}:{DB_PASSWORD}@{DB_URL}:{DB_PORT}/{DB_NAME}")
```

• [10]:

```
# Initialize LLM
llm_db = LLM(
    api_key="YOUR-API-KEY",
    api_base = "https://us-south.ml.cloud.ibm.com",
    model="watsonx/meta-llama/llama-3-3-70b-instruct",
    params={
        "decoding_method": "greedy",
        "max_new_tokens": 15000,
        "temperature": 0,
        "repetition_penalty": 1.05
    }
)
```

- Anda juga dapat membuat API key pada *homepage*. Anda dapat menyimpan *api_key* pada *local machine* anda untuk penggunaan selanjutnya.



Step 7: Run the notebook

Tahap 7: Jalankan *notebook* tersebut.

- Ikuti setiap cells yang terdapat pada *notebook* dan jalankan *agent* untuk melakukan analisis sales. Beberapa query telah diberikan dan dijadikan “*comment*” untuk dilakukan pengetesan pada hasilnya.

```
#Example Question 1: sales of Xtralife for last 3 months
#Example Question 2: Percentage change of monthly sales of Xtralife every month
user_question = input("Enter question: ")

query_agent=create_dbagent()
generation_task = create_generation_task(query_agent, user_question)

generation_crew = Crew(
    agents=[query_agent],
    tasks=[generation_task],
    verbose=True
)

crew_output = generation_crew.kickoff()
print ("Answer:..... \n" ,crew_output)
```

Enter question: Percentage change of monthly sales of Xtralife every month

 **Congratulations! You have successfully created an AI-powered Sales analyzer Agent powered by watsonx.ai and crew.ai. Happy Coding! 🎯**